

Diego Atencia

Machine Learning Engineer — Banking Data & GenAI

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SUMMARY

Mathematician (UAM) turned ML engineer with 3+ years across banking data and AI: regulatory data pipelines (Oracle, T-SQL), IRB credit-risk modeling (LGD), and production GenAI (LoRA fine-tuning, RAG, AWS). Open-source contributor to LLM training infrastructure (Liger-Kernel, TorchRL).

EXPERIENCE

[1] Nov 2023 – Present
Madrid

- Built and validated LGD (Loss Given Default) credit-risk models for IRB portfolios, ensuring regulatory compliance.
- Developed Power Automate/Copilot Studio workflows integrating LGD model outputs into structured Word reporting pipelines.

SQL & PL/SQL Developer Oct 2022 – Nov 2023
Álamo Consulting — Madrid

- Owned regulatory reporting pipelines (FINREP, COREP, SIRBE/CIRBE) for an external banking entity using Oracle and T-SQL.
- Built scalable transformations within a large regulatory codebase, ensuring accurate and timely data delivery.

OPEN-SOURCE CONTRIBUTIONS

Liger-Kernel — github.com/linkedin/Liger-Kernel PR #1128 (under review)
Production CUDA kernel library for LLM training, used at scale across industry

- Debugging a distributed-training bug in **LigerTiledSwiGLUMLP**: FSDP's post-backward reshard hooks were triggered prematurely inside a tiling loop, corrupting parameter shards across GPUs.
- Adapting axolotl's gradient-accumulator approach as the fix.

TorchRL — github.com/pytorch/rl Discussion #3538
PyTorch RL library (Meta)

- Proposing and implementing MC-PILCO: a model-based RL algorithm using Gaussian Process world models and Monte Carlo rollouts.

LEANN Issue #41
• Proposed a hybrid BM25+RRF retrieval pipeline over the neural-only architecture to improve recall on sparse queries.

SELECTED PROJECTS

regllm.xyz — GenAI app for Spanish banking regulation Q&A Production, 2026
• Fine-tuned Qwen2.5-7B with LoRA; RAG pipeline; full-stack deployment on AWS (ECS Fargate, RDS/pgvector, ALB).
• Built and shipped end-to-end; serving real user queries in early access (~30 to date).

Hand Prosthesis Control with Deep RL — blog M.Sc. thesis
• Multimodal RL agent (PPO/DDPG) integrating stereoscopic vision and sensor data for real-time prosthetic hand control in MuJoCo simulation.

EDUCATION

M.Sc. in Artificial Intelligence (8.67/10) Sep 2024 – Oct 2025
International University of Valencia
Thesis: *Optimizing low-cost hand prosthesis control using Deep Reinforcement Learning*, multimodal architecture integrating stereoscopic vision and tactile sensors.

B.Sc. in Mathematics Sep 2018 – Jun 2023
Autonomous University of Madrid (UAM)

Final thesis: Weak solutions to Navier–Stokes equations (9/10). Founder of university AI student association (70+ members). Rugby team member.

TECHNICAL SKILLS

Languages	Python (expert), C/C++ (strong), SQL (expert)
Systems & Cloud	AWS, Docker, Linux, Git/GitHub, CI/CD principles, distributed training (FSDP/DDP), CUDA kernels
Databases	Oracle, PostgreSQL, ChromaDB (vector), SAS
ML / GenAI	PyTorch, TensorFlow, Scikit-learn, HuggingFace, LoRA/QLoRA, RAG pipelines
CS Fundamentals	Data structures & algorithms, OOP, distributed systems, debugging complex systems, code review

LANGUAGES

Spanish (native) English (fluent, professional working proficiency) Mandarin Chinese (learning)